

Audio lecture four. Diabetes in pregnancy.

Companion reading handout for simultaneous listening.

This lecture is based mainly on Gabbe, chapter 40, Diabetes Mellitus Complicating Pregnancy, Israeli Position Papers number 16 and 17 on gestational diabetes and pregestational diabetes, and the local gestational diabetes mellitus protocol.

The deepest idea

Let us begin with the deepest idea.

Pregnancy is a planned metabolic stress test. Early pregnancy stores energy. Late pregnancy resists insulin so that nutrients remain available for the fetus. In most women, the pancreas compensates. In women with limited beta-cell reserve, obesity-related insulin resistance, type 1 diabetes, type 2 diabetes, or vascular complications, this normal pregnancy physiology becomes disease.

That is why diabetes in pregnancy is not only about glucose numbers. It is about placental growth, fetal fuel exposure, fetal insulin, oxygen consumption, vascular disease, organogenesis, and long-term metabolic programming.

Normal pregnancy physiology

Let us build from normal pregnancy physiology.

Gabbe describes normal pregnancy as a diabetogenic state, especially in late gestation. Postprandial glucose tends to rise more, and the maternal insulin response rises as well. Early pregnancy is different. Early pregnancy can be relatively anabolic, with increased maternal fat storage and sometimes lower insulin requirements in women with type 1 diabetes. Later, placental and maternal hormones, inflammatory mediators, and adipose-derived signals reduce insulin sensitivity.

The fasting glucose level in pregnancy is lower than outside pregnancy, often by about **10 to 15 milligrams per deciliter**. Why? Partly because of plasma volume expansion, and partly because the fetus and placenta continuously use glucose. At the same time, fasting hepatic glucose production actually increases in late pregnancy. This sounds paradoxical, but it makes sense. The mother is more insulin resistant, so the liver produces more glucose despite higher insulin levels. The placenta and fetus consume glucose, keeping fasting levels lower.

The clinical result is crucial. Pregnancy is not a state where we accept ordinary nonpregnant fasting values. A fasting value that looks only mildly abnormal outside

pregnancy may be meaningful in pregnancy.

Fetal effects

Now let us discuss fetal effects.

Glucose crosses the placenta by facilitated diffusion. Maternal insulin does not cross in meaningful amounts, but maternal glucose does. The fetus responds with its own insulin. Fetal insulin is a growth hormone. It promotes fat deposition, organ growth, and macrosomia.

This is the Pedersen logic: maternal hyperglycemia leads to fetal hyperglycemia, which leads to fetal hyperinsulinemia, which leads to overgrowth and neonatal metabolic problems.

But glucose is not the whole story. Gabbe emphasizes that lipid and amino acid metabolism also matter. Women with obesity or gestational diabetes often have altered free fatty acids and triglycerides, which contribute to fetal adiposity. So fetal overgrowth is a nutrient environment problem, not simply a sugar problem.

Now think about oxygen. Hyperinsulinemia increases fetal metabolic rate and oxygen consumption. A large fetus in a diabetic environment can be relatively hypoxic. This helps explain why stillbirth in poorly controlled pregestational diabetes is linked to chronic intrauterine hypoxia, fetal erythropoiesis, lactic acidemia, and placental dysfunction. It is also why diabetes is not benign just because the fetus is large.

Physiology summary

Pause and summarize.

Normal pregnancy lowers fasting glucose but increases insulin resistance in late gestation. Diabetes turns this normal adaptation into fetal overnutrition, fetal hyperinsulinemia, excess oxygen demand, macrosomia, neonatal hypoglycemia, and in severe cases, stillbirth.

Classification

Now we transition to classification.

Gestational diabetes mellitus, or G D M, is glucose intolerance or hyperglycemia first recognized during pregnancy, when it is not clearly overt pregestational diabetes. Israeli Position Paper number 16 defines it as glucose intolerance that appears or is first diagnosed during pregnancy. The local protocol divides it into G D M A 1, controlled by diet, and G D M A 2, requiring pharmacologic therapy.

White Class A 1 corresponds to diet-controlled gestational diabetes. White Class A 2 corresponds to gestational diabetes requiring medication, such as insulin or oral hypoglycemic therapy.

Pregestational diabetes is different. Israeli Position Paper number 17 defines it as diabetes diagnosed before pregnancy. This includes type 1 and type 2 diabetes and is associated with higher risks, especially when there is target organ damage or poor glycemic control.

The White classification for pregestational diabetes is old but still testable because it links maternal disease duration and vascular disease to risk.

White classification

- Class B means onset after age 20, duration less than 10 years, and no vascular disease.
- Class C means onset between ages 10 and 19, or duration 10 to 19 years, without vascular disease.
- Class D means disease for 20 years or more, onset before age 10, or benign retinopathy.
- Class R means proliferative retinopathy.
- Class F means nephropathy, that is, underlying renal disease. Gabbe defines Class F as reduced creatinine clearance or proteinuria of at least 500 milligrams in 24 hours during the first 20 weeks of gestation. This is a common exam answer.
- Class T means renal transplant.

The point of the White classification is not nostalgia. It reminds you that diabetes risk is not only about glucose today. It is about vessels. Retinal vessels, renal vessels, placental vessels, and coronary vessels.

Classification summary

Pause here after classification.

The short summary is this. G D M A 1 is diet controlled. G D M A 2 needs medication. Pregestational diabetes is higher risk, especially when duration is long or vascular disease is present. White Class F means nephropathy, and that detail tends to appear on exams.

Preconception care

Now let us address preconception care.

The most dangerous fetal exposure in pregestational diabetes often happens before the first prenatal visit. Cardiac, renal, and central nervous system development occur

early, especially during the first 7 weeks. Hyperglycemia during organogenesis increases congenital anomaly risk and miscarriage risk.

Gabbe emphasizes that preconception counseling and glycemic optimization reduce malformations. Elevated first-trimester hemoglobin A 1 c correlates with congenital anomalies. The goal is not merely to react once pregnancy is confirmed. The goal is to enter pregnancy already controlled.

Preconception care includes hemoglobin A 1 c assessment, medication review, folic acid, renal evaluation, retinal examination, blood pressure control, and contraception until control is acceptable. Israeli Position Paper number 17 explicitly recommends discontinuation of A C E inhibitors and A R B medications.

A C E inhibitors and A R Bs

Let us dwell on A C E inhibitors and A R Bs.

Angiotensin converting enzyme inhibitors, or A C E inhibitors, and angiotensin receptor blockers, or A R Bs, are valuable renoprotective drugs outside pregnancy. But in pregnancy they are contraindicated. They can cause fetal renal dysgenesis, proximal tubular dysgenesis, oligohydramnios, pulmonary hypoplasia, growth restriction, and fetal or neonatal renal failure. The logic is direct. These drugs interfere with the fetal renin angiotensin system, which is important for renal perfusion and development. Damaged fetal kidneys make less urine. Less urine means oligohydramnios. Oligohydramnios can impair lung development.

So the medication that protects the mother's kidney outside pregnancy can injure the fetal kidney inside pregnancy.

Diabetic nephropathy

Now let us talk about diabetic nephropathy.

Gabbe defines overt diabetic nephropathy by persistent proteinuria in the absence of infection or other urinary disease. In pregnancy, Class F includes proteinuria of at least **500 milligrams per 24 hours before 20 weeks** or reduced creatinine clearance. Severe early proteinuria, especially more than **3 grams per day**, and serum creatinine above **1.5 milligrams per deciliter** predict worse outcomes, including preterm birth, low birth weight, and preeclampsia.

Pregnancy normally increases glomerular filtration rate. Women with diabetic nephropathy often do not mount that normal increase. Proteinuria often worsens during gestation, sometimes into the nephrotic range, and then may improve postpartum. This creates a diagnostic challenge: is the patient developing preeclampsia, or is nephropathy progressing as expected? Baseline 24 hour urine

testing is therefore not administrative busywork. It is the reference point that allows interpretation later.

Blood pressure control is crucial. Gabbe notes that calcium channel blockers have renoprotective effects similar to A C E inhibitors and do not appear teratogenic, making them a favored option for hypertension in pregnant women with diabetic nephropathy. Labetalol and hydralazine are other options.

This is another likely exam point: **diabetic nephropathy in pregnancy, avoid A C E inhibitors and A R Bs, consider calcium channel blockers.**

Retinopathy

Now retinopathy.

Pregnancy can worsen existing diabetic retinopathy, especially when retinopathy is present at conception, diabetes duration is long, glycemic control is poor or rapidly normalized, and hypertension is present. This does not mean good control is bad. It means that rapid metabolic shifts in a vulnerable retina can unmask progression. Women planning pregnancy should have a comprehensive eye exam and treatment before conception when needed.

Complications summary

Pause here and connect the complications.

The major pregestational diabetes complications are congenital anomaly risk from early hyperglycemia, nephropathy, retinopathy, hypertension, preeclampsia, fetal growth problems, and stillbirth risk when control or vascular status is poor. This is why preconception care and baseline organ assessment matter.

Screening and diagnosis of gestational diabetes

Now let us transition to screening and diagnosis of gestational diabetes.

Most practices in Israel and the United States use a two-step approach. First, a 50 gram glucose challenge test, or G C T, at **24 to 28 weeks**. No fasting is required. If the one-hour plasma glucose is **140 milligrams per deciliter or higher**, the patient proceeds to a diagnostic 100 gram oral glucose tolerance test, or O G T T.

In the 100 gram test, after an 8 to 14 hour fast, plasma glucose is checked fasting, then at one, two, and three hours. The Carpenter and Coustan thresholds are **95, 180, 155, and 140 milligrams per deciliter**. Diagnosis is usually made when two or more values are abnormal.

Israeli Position Paper number 16 and the local protocol add nuance. A G C T above **200** can define gestational diabetes without further testing in many settings. The local protocol also considers a G C T above 200 plus fasting glucose above **125**, or random glucose **200 or above**, as diagnostic without additional confirmation.

One abnormal value on O G T T is not ignored. If risk factors exist, such as polyhydramnios, maternal age over 40, estimated fetal weight over 4 kilograms, or prior G D M, the local protocol supports defining it as diabetes. Without risk factors, further dietary glucose profiling or repeat testing may be used.

Diagnosis summary

Pause after diagnosis.

For the test, keep the sequence simple. Screen at 24 to 28 weeks with a 50 gram G C T. If abnormal, diagnose with a 100 gram O G T T using Carpenter and Coustan thresholds. A very high G C T, especially above 200, can be diagnostic in many local pathways.

Treatment

Now treatment.

Treatment begins with nutrition and physical activity. This is not because G D M is mild. It is because many cases are driven by postprandial physiology and can respond to meal composition, carbohydrate distribution, and exercise-mediated glucose uptake.

Glucose targets commonly include fasting below **95 milligrams per deciliter**, one hour after meals below **140**, and two hours after meals below **120**. If diet and activity do not achieve targets within a few weeks, pharmacologic therapy is added.

Insulin remains the standard medication because it does not cross the placenta in the way oral agents do and can be titrated precisely. Metformin can be used according to Israeli guidance; it crosses the placenta and fetal levels may approach maternal levels, so long-term offspring questions remain. Glyburide, or glibenclamide, is another option in some guidance, but practice varies.

Treatment summary

Pause after treatment.

The management ladder is nutrition and activity first, then medication if targets are not met. Insulin is the reference medication. Metformin is used in some guidance, but it crosses the placenta, so counseling should be explicit.

Fetal surveillance and delivery

Now fetal surveillance and delivery.

In uncomplicated diet-controlled G D M, Gabbe does not routinely institute antepartum fetal heart rate testing, because stillbirth risk appears low. Israeli Position Paper number 16 says fetal monitoring can be considered from **32 to 34 weeks** depending on risk factors such as medication need, poor control, obstetric history, or maternal disease.

For pregestational diabetes, especially with vascular disease or poor control, surveillance is more intensive. Growth ultrasound is important because diabetes can cause macrosomia, but vascular disease can cause fetal growth restriction. This is the paradox of diabetes: without vasculopathy, the fetus may be overgrown; with vasculopathy, the placenta may underperform.

Delivery timing depends on diabetes type, control, vascular disease, fetal growth, and complications. The local protocol suggests **39 weeks** for controlled pregestational diabetes without complications. For G D M, well-controlled cases generally should not be delivered early solely for the diagnosis if fetal weight is appropriate and no other risk factors exist. If estimated fetal weight is high, elective cesarean thresholds vary. Israeli Position Paper number 16 uses over **4250 grams** for gestational diabetes, while the local protocol uses over **4000 grams** in diabetes as its institutional position.

Postpartum

Finally, postpartum.

G D M is not over at delivery. It is a marker of future type 2 diabetes risk. The local protocol recommends stopping pharmacologic treatment after delivery for gestational diabetes or prediabetes states, and performing a 75 gram O G T T at **4 to 12 weeks postpartum**. Gabbe emphasizes that women with G D M have a markedly increased long-term risk of type 2 diabetes, and repeat testing should continue over time.

Surveillance, delivery, and postpartum summary

Pause and summarize surveillance, delivery, and postpartum care.

Diet-controlled G D M usually has less need for intensive antepartum testing than pregestational diabetes. Poor control, medication use, vascular disease, or abnormal growth raises concern. After delivery, G D M becomes a long-term diabetes prevention issue, not a closed pregnancy chapter.

Final synthesis

Now final synthesis.

1. **First, pregnancy is a metabolic stress test.** Late gestation insulin resistance is normal, but G D M appears when beta-cell reserve cannot compensate.
2. **Second, fetal risk is driven by fetal insulin.** Maternal glucose crosses the placenta; fetal insulin creates growth, adiposity, neonatal hypoglycemia, and increased oxygen demand.
3. **Third, pregestational diabetes is dangerous early.** Organogenesis happens before many patients present, so preconception control is critical.
4. **Fourth, Class F means nephropathy.** Proteinuria at least 500 milligrams per 24 hours before 20 weeks is a key definition. Avoid A C E inhibitors and A R Bs. Calcium channel blockers are favored by Gabbe for hypertension with diabetic nephropathy.
5. **Finally, G D M A 1 means diet-controlled gestational diabetes.** It still matters because it predicts future maternal diabetes and may shape offspring metabolic risk.

End of lecture four. Pause here before continuing to early pregnancy loss, ectopic pregnancy, and recurrent pregnancy loss.